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ChE 311 Quiz 5, Fall 2019

Closed book and closed notes, 15 minutes, 5 points

1L = 1×10⁻³ m³

cp =

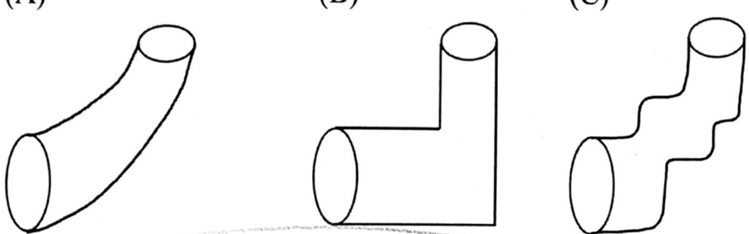
1. Choose the non-zero quantities from the following list.

(A) $\overline{V_z'}$ (B) $\overline{V_z}\,\overline{V_z}$ (C) $\overline{V_x}\overline{V_z'}$ (D) $\overline{(V_z')^2}$ (E) $\overline{V_x' V_z'}$

$\overline{V_x}\,\overline{V_z'} = 0$

Answer: D, E

2. Which one of the following pipe bend needs a bigger force to support it to be stationary? The bending angle from the inlet to outlet, entrance pressure, exit pressure, flow rate, and type of fluid are the same. Neglect the volume difference of the fluid in the pipe.



(D) The force is the same for all three pipes.

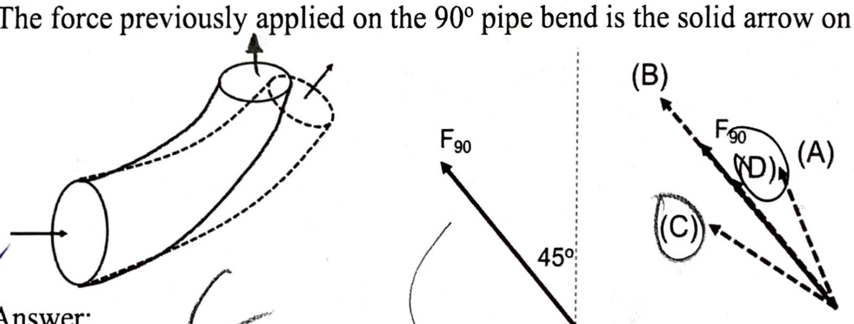
Answer: D

$6366 \frac{kg}{m^3}\frac{L}{min} \cdot \frac{1}{cp} \cdot \frac{1}{m}$

$6366 \frac{kg\,L}{m^3\,min} \cdot \frac{1}{cp} \cdot \frac{1}{m} \Big| \frac{1\times10^{-3}\,m^3}{1\,L}$

$0.1061 \frac{kg}{m\,s\cdot cp} \Big| \frac{1\,cp}{1\times10^{-3}\,kg/m\cdot s}$

3. For a pipe bend, if we change the bending angle from 90° to 60°, please choose the one best represents the force applied on the 60° pipe bend to balance the force by the fluid on the pipe. The force previously applied on the 90° pipe bend is the solid arrow on the figure.



F_y ↑ less

Answer: C

4. For a 90° pipe bend, the pipe diameter is 20 cm. The flow rate is 100 L/min. Please estimate the Reynolds number for water flow through the pipe. Consider water density 1000 kg/m³ and viscosity 1 cp at room temperature.

1000 g/m³ ρ = 1 g/mL

(A) 10 (B) 10^2 (C) 10^3 (D) 10^4 (E) 10^5

$Re = \frac{\rho \langle v \rangle d}{\mu}$

$\langle v \rangle = \frac{Q}{A}$

Answer: D

$Re = \frac{\rho V L}{\mu} = \frac{\rho \frac{Q}{A} D}{\mu}$ $V = Q/A$

5. For above flow in problem 4, what is $\langle V^2\rangle/\langle V\rangle$ at the inlet?

$Re \gg 0$

$\frac{\langle v^2\rangle}{\langle v\rangle} = \langle v\rangle = \frac{Q}{A} = \frac{100\,L/min}{\frac{1}{4}\pi D^2}$

$\frac{\langle v^2\rangle}{\langle v\rangle} = \langle v\rangle = Q/A$

$= \frac{100\,L/min}{\frac{1}{4}\pi(20\,cm)^2}$

$= 0.3183 \frac{L}{min} \cdot \frac{1}{cm^2} \Big| \frac{10^3\,cm^3}{1\,L}$

$= 318.3\,cm/min$

$= \rho \frac{Q}{\frac{1}{4}\pi D^2} \frac{D}{\mu}$

$= \frac{\rho Q}{\frac{1}{4}\pi D} \cdot \frac{1}{\mu}$

$= (1000\,kg/m^3)(100\,\frac{L}{min})$